

SIMONE ROMANO

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sroman0.github.io ◇ **Willing to relocate**

PROFILE

Cybersecurity Engineering master's student focused on Linux security, vulnerability research, digital forensics, and secure systems. Experienced in turning research questions into reproducible experiments, from Linux kernel exploitability analysis to embedded-system emulation and security-oriented machine learning. Master's thesis intern at Rakuten Mobile in Tokyo since July 2026.

EDUCATION

Politecnico di Torino 2024 – Present
M.Sc. in Cybersecurity Engineering Turin, Italy
English-taught programme compatible with the European Cyber Security Organisation (ECSO) curriculum standards.
Current weighted average: **28.98/30 over 98 graded CFU**
Master's thesis work on Linux runtime security and eBPF-based monitoring, in collaboration with Rakuten Mobile.

Università degli Studi del Sannio 2019 – June 2024
B.Sc. in Computer Engineering, final grade: **100/110** Benevento, Italy
Thesis: *Design Solutions and Their Impact on Smart-Contract Execution Costs*.
<https://zenodo.org/records/20776825>

M.SC. ACADEMIC RECORD

Grades from the official Politecnico di Torino record dated 10 July 2026. The weighted average counts 30 cum laude as 30.

30/30 cum laude: Information Systems Security (8 CFU); Cryptography (6 CFU); Security Verification and Testing (8 CFU).

30/30: Web Applications (8 CFU); Networks & Cloud Technologies and Security (12 CFU); Computer Forensics and Cybercrime Analysis (8 CFU).

29/30: Computer Architectures and Operating Systems (8 CFU); AI and Cybersecurity (6 CFU); Cybersecurity Laws and Regulations (6 CFU).

28/30: Hardware & Wireless Security (12 CFU).

27/30: Advanced Information Systems Security (8 CFU).

26/30: Machine Learning for Networking (8 CFU).

EXPERIENCE & LEADERSHIP

Rakuten Mobile July 2026 – Present
Master's Thesis Intern Tokyo, Japan

- Developing the process-monitoring path of an eBPF-based Linux runtime-security tool, covering process creation, execution, and termination, with structured event delivery to Go userspace.
- Validated BTF, CO-RE, BPF LSM, ring-buffer, and stack capabilities on Rocky Linux 8.10 rather than relying only on its reported upstream kernel version.

IEEE-Eta Kappa Nu, Mu Nu Chapter 2025 – 2026
Events Area Member Politecnico di Torino

- Supported technical and professional-development events through logistics, team coordination, budgeting, and sponsor engagement.

AWARDS & RECOGNITION

1st place, IEEE-HKN Global Hackathon 2025: team SpaghettiOverflow, representing the Mu Nu Chapter.
57th / 2,000 teams, Reply AI Agents Challenge 2026.
88th / 2,000 teams, Reply Hack The Code Challenge 2025: team FARES.
IEEE-Eta Kappa Nu Honor Society member: Mu Nu Chapter, Politecnico di Torino.

SELECTED PROJECTS & RESEARCH

Linux eBPF Vulnerability Research

2026

Independent Security Research

Politecnico di Torino

- Reproduced four Linux eBPF verifier vulnerabilities in isolated Buildroot/QEMU environments and traced each issue from the incorrect verifier decision to its concrete security impact.
- Developed two controlled local privilege-escalation chains for CVE-2023-39191 and CVE-2024-58100; documented assumptions, hardening constraints, and reliability boundaries.

SSH Shell Attack Classification

2025

Machine Learning for Security

Four-person team

- Analyzed 230,000 honeypot sessions and 90,026 encoded shell scripts; compared supervised models, clustering, and BERT for multi-label attacker-intent classification while documenting class-imbalance limits.

Secure Timeout System for NXP S32K358EVB

2024 – 2025

Embedded Systems

Five-person team

- Extended QEMU with an automotive-board model, integrated FreeRTOS, UART, interrupts, and periodic timers, and produced a reproducible build and GDB debugging workflow.

Shattered Bytes

2026

Digital Forensics & Learning Design

Individual project

- Designed and developed a six-part browser game that teaches manual data carving, evidence validation, uncertainty reporting, and defensible forensic reasoning.

Smart-Contract Design and Gas Costs

2024

Bachelor's Thesis Research

Individual project

- Studied 1,010 Ethereum functions and 18 architectural patterns; six patterns showed a statistically significant association with higher execution costs using Fisher's exact test.

TECHNICAL SKILLS

Programming: C, C++, Python, Go, Java, Bash, Solidity, SQL, ProVerif

Security & systems: Linux, eBPF, vulnerability analysis, exploit development, digital forensics, network security, OpenSSL

Platforms & tools: Git, QEMU, Buildroot, FreeRTOS, GDB, LaTeX

Machine learning: scikit-learn, PyTorch, text classification, clustering, BERT

LANGUAGES & CERTIFICATIONS

Italian: Native

English: Cambridge B2 First, credential ID 86238oAXK

Additional training: GNU/Linux Advanced, Politecnico di Torino